The education-obesity link across countries over time

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Tutorial group: 6.1

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Part 1 - Identify a Social Problem

The social problem we have identified is the link between the BMI Index and educational attainment level, bridging two data sets from Eurostat to compare and identify any correlation between the two factors.

1.1 Describe the Social Problem

The relevance of these two variables, BMI and education levels, and how they are correlated with each other lies in the fact that obesity comes with serious health risks, such as diabetes and cardiovascular issues, which are often more common among people with lower educational attainment (World Health Organization, 2022; European Commission, 2024). This then results in the burden of health risks being disproportionately put on the lower educated class of society, and further increases social inequality issues. Beyond the health risks themselves, education has a clear impact on the ability of a person to maintain a healthy lifestyle, as higher education often comes with higher income, better access to higher quality food and so on. This means that addressing obesity cannot be done through increasing awareness alone, and that broader structural interventions, such as food pricing policies or workplace conditions need to be considered.

The fact we are looking at the European aspect of this issue also adds to its relevance. By looking at multiple countries over time, it becomes possible to assess whether this education gap in obesity rates is a universal pattern across Europe or whether certain countries have managed to narrow it. This kind of cross country comparison is useful for understanding what policy approaches may actually work in practice, especially since educational inequalities in obesity have been documented across Europe in recent research (World Health Organization, 2022; European Commission, 2024).

While the existence of an educational gradient in obesity is well established in the literature (World Health Organization, 2022; European Commission, 2024), far less is settled about how this gap behaves over time and whether it is narrowing or widening across different European countries. Much existing work documents the association at a single point in time or within individual countries, which leaves the question of long-term, cross country trends comparatively underexplored. This is the gap our project addresses: by tracking obesity rates by education level across multiple European countries and survey years, we aim to establish whether the education obesity divide is a stable, universal pattern or one that some countries have managed to reduce.

Part 2 - Data Sourcing

2.1 Load in the data

Both datasets are published by Eurostat. We downloaded them once and saved them as CSV files in the `data/` folder of our repository, which we read into R with `read_csv()`. Using saved copies rather than querying Eurostat live means the report always knits from the same data and does not depend on a network connection or on the Eurostat servers being reachable. The files are small, so storing them in the repository keeps the project self-contained and reproducible: anyone who clones it can knit the report without extra setup.

```r
# Load the saved datasets from our repository
bmi_edu <- read_csv("data/bmi_edu.csv")
obesity_trend <- read_csv("data/obesity_trend.csv")
```

2.2 Provide a short summary of the dataset(s)

```r
head(bmi_edu)
#> # A tibble: 6 x 9
#>   freq  unit  bmi        isced11 sex   age   geo   TIME_PERIOD values
#>   <chr> <chr> <chr>      <chr>   <chr> <chr> <chr>       <dbl>  <dbl>
#> 1 A     PC    BMI18P5-24 ED0-2   F     TOTAL AT           2014   45.6
#> 2 A     PC    BMI18P5-24 ED0-2   F     TOTAL AT           2019   40.9
#> 3 A     PC    BMI18P5-24 ED0-2   F     TOTAL BE           2014   45.7
#> 4 A     PC    BMI18P5-24 ED0-2   F     TOTAL BE           2019   41
#> 5 A     PC    BMI18P5-24 ED0-2   F     TOTAL BG           2014   44.3
#> 6 A     PC    BMI18P5-24 ED0-2   F     TOTAL BG           2019   48.1
head(obesity_trend)
#> # A tibble: 6 x 6
#>   freq  unit  bmi     geo   TIME_PERIOD values
#>   <chr> <chr> <chr>   <chr>       <dbl>  <dbl>
#> 1 A     PC    BMI25-29 AL          2022   44.8
#> 2 A     PC    BMI25-29 AT          2008   36.5
#> 3 A     PC    BMI25-29 AT          2014   33.3
#> 4 A     PC    BMI25-29 AT          2017   35
#> 5 A     PC    BMI25-29 AT          2019   35
#> 6 A     PC    BMI25-29 AT          2022   35
```

Our first dataset, bmi_edu (Eurostat code hlth_ehis_bm1e, Body mass index (BMI) by sex, age and educational attainment level), comes from the European Health Interview Survey (EHIS), in which respondents report their own height and weight (Eurostat, n.d.-a). For each country and year it gives the share of people in each BMI category, broken down by education level, sex, and age. EHIS runs only every few years, so the data covers mainly the 2014 and 2019 waves. This is the dataset that links weight to education, so it is central to our analysis.

Our second dataset, obesity_trend (Eurostat code sdg_02_10, Obesity rate by body mass index (BMI)), is one of the EU's Sustainable Development Goal indicators, also based on self-reported health surveys (Eurostat, n.d.-b). It does not split the figures by education, but it covers many more years (2008, 2014, 2017, 2019, 2022), which makes it well suited to tracking obesity over time. It supplies the "across countries, over time" side of our research question. The data are reported by country, covering the EU member states together with several additional European countries, each shown as a two-letter code in the geo column.

2.3 Describe the type of variables included

Table 1: Variables in the two datasets.

Variable	Type	Dataset	Meaning
freq	character	both	data frequency (A = annual)
unit	character	both	unit of measurement (PC = percentage)
bmi	character	both	BMI category (e.g. overweight, obese)
isced11	character	bmi_edu	education level (ISCED 2011 group)
sex	character	bmi_edu	sex (M, F, T = total)
age	character	bmi_edu	age group
geo	character	both	country (Eurostat code)
TIME_PERIOD	numeric	both	survey year
values	numeric	both	share of people in that category

As Table 1 shows, the variables fall into three groups. The health measure is BMI/obesity itself, the share of people in each weight category (the `bmi` and `values` columns). The socioeconomic measure is education (the `isced11` column in `bmi_edu`), which is the variable at the heart of our question. The remaining columns (`sex`, `age`, `geo` for country, and `TIME_PERIOD` for year) are background characteristics that we use to split the figures by group, place, and time.

Neither dataset comes from administrative records; both rely on surveys in which people report their own height and weight. This has two implications. First, self reported data tends to underestimate obesity slightly, since people often round their weight down. Second, the two datasets are not fully independent: the SDG indicator is partly built on the same EHIS survey as `bmi_edu`, so they are related rather than separate sources. We combined them deliberately: `bmi_edu` provides the education breakdown, while `obesity_trend` adds the longer time span. Because both are self reported survey data, our analysis can describe associations (for example, that obesity is more common among lower educated groups) but cannot show that education causes obesity.

Part 3 - Quantifying

3.1 Data cleaning

Stefan add explanation here

Cleaning Step 1: Filter and clean the first dataset

```
bmi_edu_clean <- bmi_edu %>%
  filter(
    bmi == "BMI_GE30",                            # Keep only Obese rows
    isced11 %in% c("ED0-2", "ED3_4", "ED5-8"), # Separate the 3 education groups
    sex == "T",                                 # Combined sexes
    age == "TOTAL"                            # Combined adult age group
  ) %>%
  select(geo, TIME_PERIOD, isced11, bmi_value = values) %>%
  drop_na(bmi_value)
```

and here

Cleaning Step 2: Filter and clean the second dataset

```
obesity_trend_clean <- obesity_trend %>%
  filter(
    bmi == "BMI_GE30" # Keep only Obese rows since that's what we are comparing
  ) %>%
  select(geo, TIME_PERIOD, trend_value = values) %>%
  drop_na(trend_value)
```

and here

Cleaning Step 3: Combine both datasets into one master file

```
combined_dataset <- inner_join(bmi_edu_clean, obesity_trend_clean, by = c("geo", "TIME_PERIOD"))
```

and here

3.2 Generate necessary variables

and here

Variable 1

```
# Create Variable 1: Human-readable education levels
bmi_edu_with_var1 <- combined_dataset %>%
  mutate(
    edu_level = case_when(
      isced11 == "ED0-2" ~ "Low Education",
      isced11 == "ED3_4" ~ "Medium Education",
      isced11 == "ED5-8" ~ "High Education"
    )
  )
```

```
# Reshaping before subtracting
gap_data <- bmi_edu_with_var1 %>%
  # 1. Isolate the columns we need
  select(geo, TIME_PERIOD, edu_level, bmi_value) %>%

  # 2.Flip the rows into columns
  pivot_wider(
    names_from = edu_level,
    values_from = bmi_value
  )
```

and here

Variable 2

```
# Create Variable 2: Calculate the obesity inequality gap
final_gap_data <- gap_data %>%
  mutate(
    obesity_gap = `Low Education` - `High Education`
  )
```

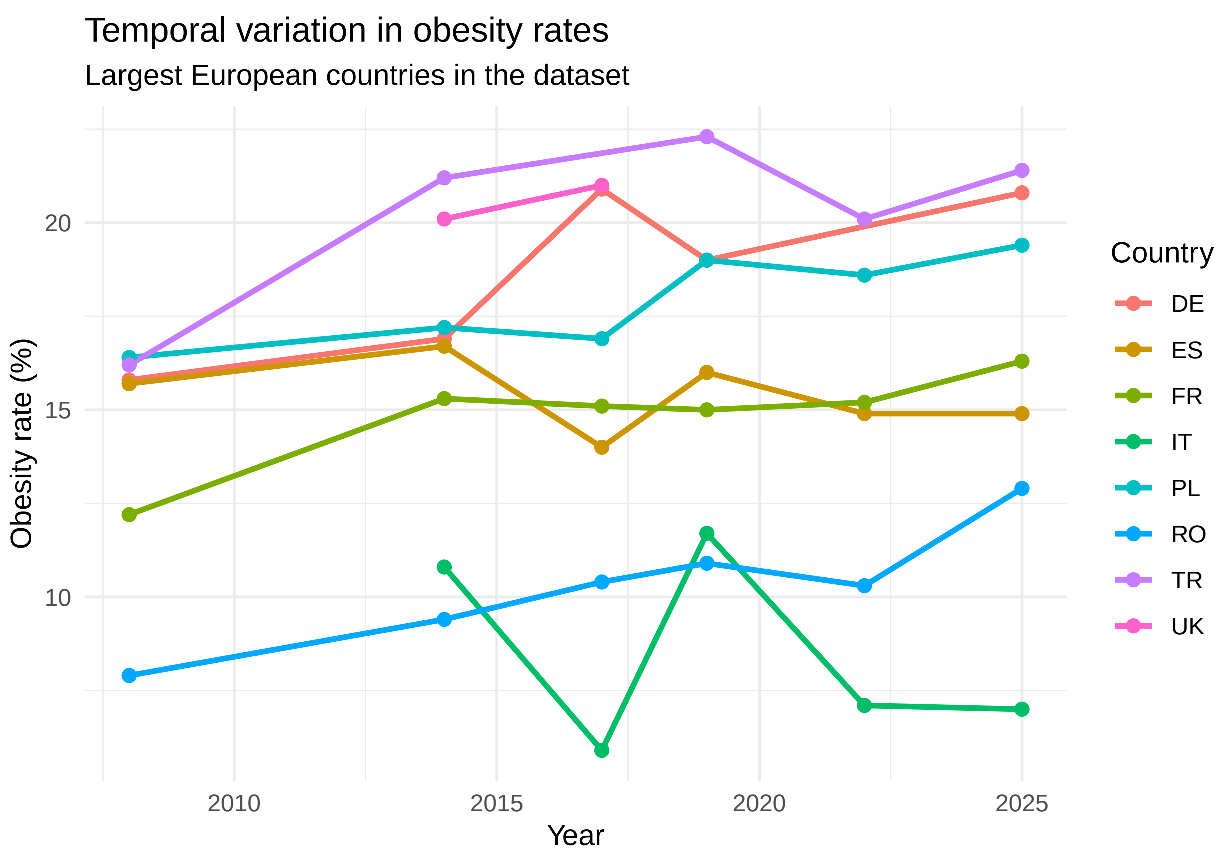
3.3 Visualize temporal variation

```
# First I make a vector of the landcodes of the countries in Europe with the largest population, becaus
largest_european_countries <- c("RU", "TR", "DE", "UK", "FR", "IT", "ES", "UA", "PL", "RO")
# RU:Russia, TR:Turkey, DE:Germany, UK:UnitedKingdom, FR:France, IT:Italy, ES:Spain, UA:Ukraine, PL:Pol

# The following code only selects the countries that are ACTUALLY in the dataset. So now if Russia, for
selected_countries <- largest_european_countries[
  largest_european_countries %in% obesity_trend_clean$geo
]

# This line shows the countries that are selected.
selected_countries
#> [1] "TR" "DE" "UK" "FR" "IT" "ES" "PL" "RO"
```

```r
obesity_trend_clean %>%
  filter(geo %in% selected_countries) %>%
  ggplot(aes(x = TIME_PERIOD, y = trend_value, color = geo, group = geo)) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
    labs(
      title = "Temporal variation in obesity rates",
      subtitle = "Largest European countries in the dataset",
      x = "Year",
      y = "Obesity rate (%)",
      color = "Country"
    ) +
    theme_minimal()
```

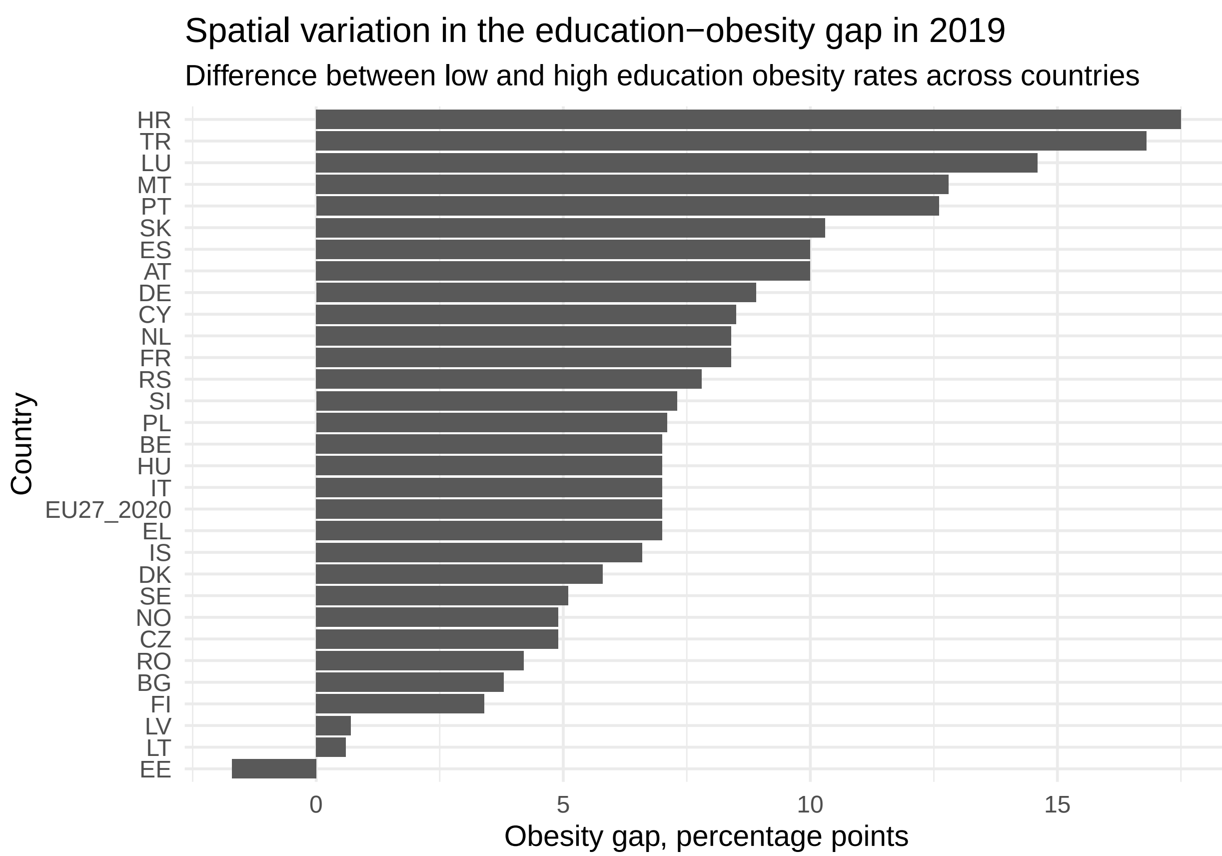



```r
# line 185: we can not use final_gap_data because otherwise we do not have the required periods since t
# Line 186: This filters only the selected countries from the dataset
# Line 187: this plots the obesity over time, x is the years on the x-axis, y is the obesity rate on th
# Lines 188-197: the codes show how the lines are shown for example the width etc, it also creates the
```

3.4 Visualize spatial variation

```r
latest_year <- max(final_gap_data$TIME_PERIOD, na.rm = TRUE)

final_gap_data %>%
  filter(TIME_PERIOD == latest_year) %>%
  ggplot(aes(x = reorder(geo, obesity_gap), y = obesity_gap)) +
  geom_col() +
  coord_flip() +
  labs(
    title = paste("Spatial variation in the education-obesity gap in", latest_year),
    subtitle = "Difference between low and high education obesity rates across countries",
    x = "Country",
    y = "Obesity gap, percentage points"
  ) +
  theme_minimal()
```




```r
# Line 208 selects the last year that we are able to use
# Line 211 filters out everything except the last year
# Line 212-221: creates the plot geom_col makes the bar plot, coord_flip flips the plot so it is nicer
```

Here you provide a description of why the plot above is relevant to your specific social problem.

This figure shows the regional differences in the education-obesity gap between countries in the most recent year for which data is available. The gap is calculated by subtracting the percentage of obese people among the low-skilled from the percentage of obese people among the highly skilled.

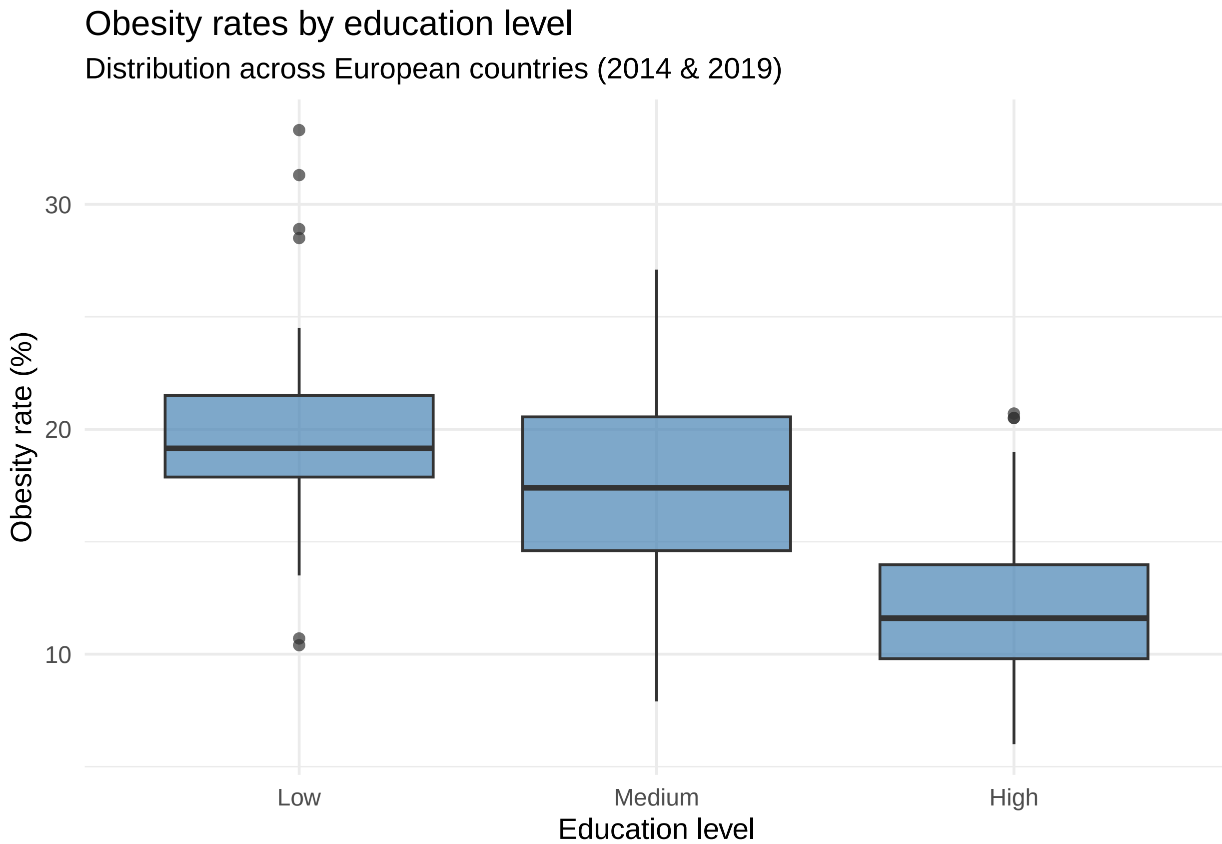
A positive value indicates that obesity is more prevalent among the low-educated. The countries are ranked according to the size of the gap, making it easier to see where education-related inequality in obesity is greatest.

3.5 Visualize sub-population variation

What is the obesity rate by country?

```r
subpop_data <- bmi_edu %>%
  filter(
    bmi == "BMI_GE30",
    isced11 %in% c("ED0-2", "ED3_4", "ED5-8"),
    sex == "T",
    age == "TOTAL"
  ) %>%
  mutate(
    edu_level = case_when(
      isced11 == "ED0-2" ~ "Low",
      isced11 == "ED3_4" ~ "Medium",
      isced11 == "ED5-8" ~ "High"
    ),
    edu_level = factor(edu_level, levels = c("Low", "Medium", "High"))
  ) %>%
  drop_na(values)

ggplot(subpop_data, aes(x = edu_level, y = values)) +
  geom_boxplot(fill = "steelblue", alpha = 0.7) +
  labs(
    title = "Obesity rates by education level",
    subtitle = "Distribution across European countries (2014 & 2019)",
    x = "Education level",
    y = "Obesity rate (%)"
  ) +
  theme_minimal()
```




```
#line 241-246 selects the data that is been used
#line 249-252 gives the three education groups understandable names
#line 254 fixes the order of education levels
#line 258-264 creates the boxplot with title and axis labels
```

The plot above is relevant because it clearly shows how a lower education level is correlated to a higher obesity rate. Here you provide a description of why the plot above is relevant to your specific social problem.

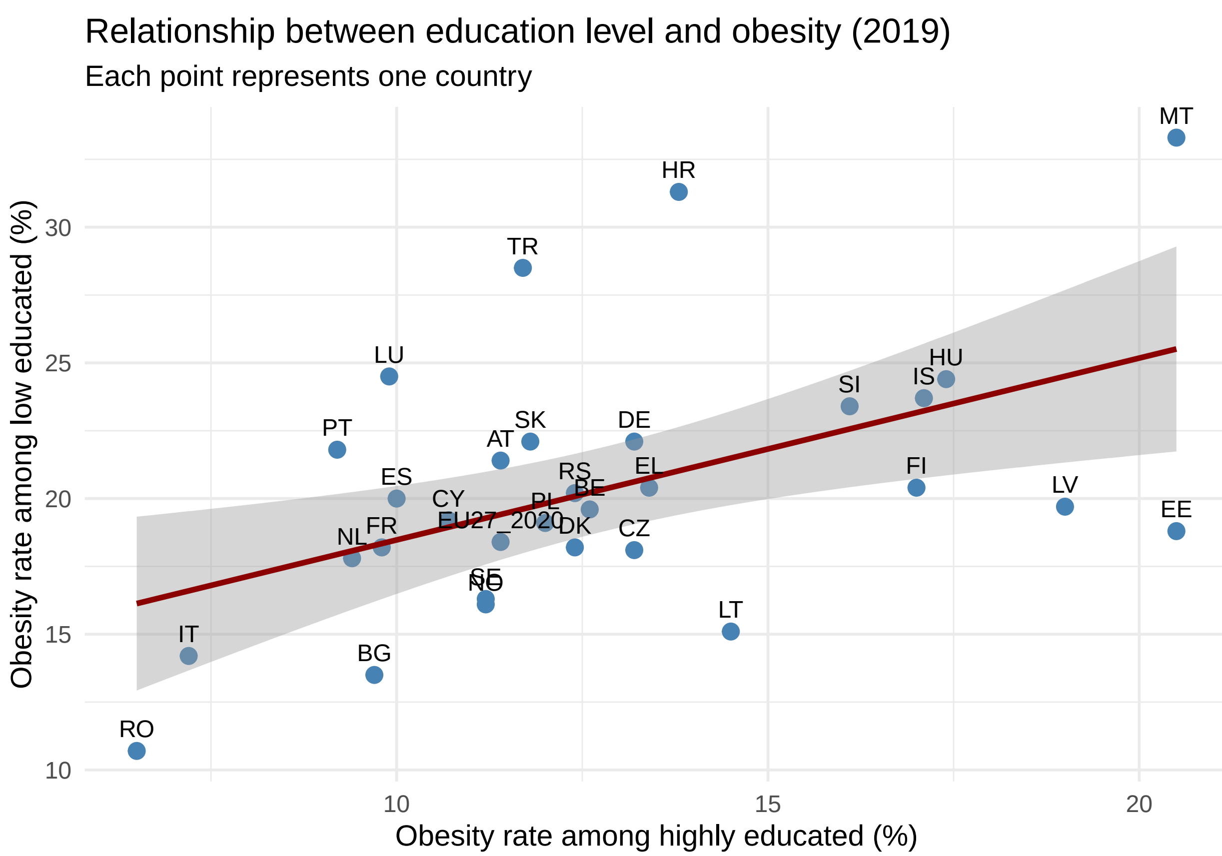
3.6 Event analysis

Analyze the relationship between two variables (BMI .

```
# Calculate average obesity rate per country per education level for the latest year
event_data <- bmi_edu_clean %>%
  filter(TIME_PERIOD == 2019) %>%
  pivot_wider(
    names_from = isced11,          # Spread education levels into separate columns
    values_from = bmi_value
  ) %>%
  rename(
    low = `ED0-2`,
```

```
    medium = `ED3_4`,
    high = `ED5-8`
  ) %>%
  drop_na()

# Scatter plot: low vs high education obesity rate per country
ggplot(event_data, aes(x = high, y = low)) +
  geom_point(size = 2.5, colour = "steelblue") +     # One dot per country
  geom_smooth(method = "lm", se = TRUE, colour = "darkred") +  # Regression line with confidence band
  geom_text(aes(label = geo), vjust = -0.8, size = 3) +          # Country labels above each dot
  labs(
    title = "Relationship between education level and obesity (2019)",
    subtitle = "Each point represents one country",
    x = "Obesity rate among highly educated (%)",
    y = "Obesity rate among low educated (%)"
  ) +
  theme_minimal()
```




```
#line 285-296 selects data that is been used
#line 299 creates the plot with high education on x and low education on y
#line 304-307 adds title and axis labels
# Print the correlation coefficient
cor.test(event_data$high, event_data$low)
#>
#>  Pearson's product-moment correlation
```

```
#>
#> data:  event_data$high and event_data$low
#> t = 3.0744, df = 29, p-value = 0.004563
#> alternative hypothesis: true correlation is not equal to 0
#> 95 percent confidence interval:
#>  0.1716049 0.7231022
#> sample estimates:
#>       cor
#> 0.4957973
```

The plot above is relevant to our social problem because it shows per country if our hypothesis is true. It also shows for which countries the theory holds more. The reason why we put the regression line with confidence band is because it clearly shows which countries are the outliers in obesity rates and which are exceptions. The plot for example shows that in Croatia the lower educated has more than double obesity rate than higher educated. This is also visible in the spatial variation. So it is clear that our graphs support each other. Here you provide a description of why the plot above is relevant to your specific social problem.

Part 4 - Discussion

4.1 Discuss your findings

Part 5 - Reproducibility

5.1 Github repository link

https://github.com/theodorkubi/ProgrammingForEcon

5.2 Reference list

European Commission. (2024). Health inequalities: Overweight and obesity. Knowledge4Policy. https://knowledge4policy.ec.europa.eu/health-promotion-knowledge-gateway/health-inequalities-5_en

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Eurostat. (n.d.-a). Body mass index (BMI) by sex, age and educational attainment level [Data set]. Retrieved June 7, 2026, from https://ec.europa.eu/eurostat/databrowser/view/hlth_ehis_bm1e

Eurostat. (n.d.-b). Obesity rate by body mass index (BMI) [Data set]. Retrieved June 7, 2026, from https://ec.europa.eu/eurostat/databrowser/view/sdg_02_10